Are the neurons of a reward-trained Ms. Pac-Man agent already clean? A preregistered neuron audit with causal steering

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1 Abstract

Question. Goodfire’s essay “Neural Networks Are Already Clean” argues that most individual neurons of trained networks have one main job. We test this claim, and whether single neurons can steer behaviour, in a deep RL agent that learned Atari Ms. Pac-Man from game reward alone. The agent is a public Sample Factory APPO agent with a Nature-CNN torso, trained for 2B environment steps.

Setup. - We re-implement the agent in PyTorch, bit-exact against the original stack (max logit difference 1.1e-05). - We decode exact game state from the console RAM. This includes two frightened-mode addresses that we identify and validate (97.1% agreement with sprite colours). - We label every unit against 40 ground-truth features.

Neuron audit (preregistered H1). On held-out games, the fraction of alive fc neurons with one main job is:

unit set	one-job fraction
agent	0.8%
random directions	0.2%
untrained network	9.4%

H1 is supported. An exploratory conjunction analysis shows the neurons are better described as place × heading cells (median R^2 0.20).

Steering (preregistered H2). Two candidates were frozen before any held-out data. On held-out games, paired by seed: - Zero-ablating fc neuron 453 changes ghosts eaten per power pill from – to – (pending the preregistered test against 99 matched random-unit nulls). - Setting conv3 channel 18 at one spatial cell changes the rate of approaching that maze location from – to – (pending).

We report all preregistered tests, deviations, controls and negative results.

2 1. Introduction

Motivation. Mechanistic interpretability asks whether the internal units of a network can be read as human-understandable variables and whether manipulating them changes behaviour in the predicted way.

The claim we test. Goodfire recently argued that most neuron poles of large vision and language models “have one main job”, based on top-activating examples, feature visualisation and steering (Cammarata/Goodfire 2026).

Why RL is a sharp test. Reinforcement-learning agents trained from reward alone offer two advantages: - the environment state is fully known (for Atari, the console RAM); - behaviour can be measured causally in closed loop.

Gap in prior work. Prior interpretability work on Ms. Pac-Man used: - input saliency (Greydanus et al. 2018); - attention maps (Mott et al. 2019); - NMF concepts (Bekkemoen & Langseth 2025); - a sparse autoencoder without interventions (DuPlessie 2024).

Single-channel steering of RL agents has been shown for a Procgen maze agent (Mini, Turner et al. 2023) and for Sokoban planners (Taufeeque et al. 2024; Bush et al. 2025). To our knowledge, no published work has steered individual neurons of a reward-trained Ms. Pac-Man agent.

Contributions. 1. A bit-exact, dependency-light re-implementation of the public Sample Factory Ms. Pac-Man agents with a golden fidelity test. 2. RAM ground truth for Ms. Pac-Man, including the frightened-mode timer `ram[116]` and per-ghost edible bits (bit 7 of `ram[1+i]`), and a maze lattice graph. 3. A preregistered Goodfire-style audit of every fc neuron against three controls (H1). 4. Preregistered causal steering tests with matched random-unit nulls on held-out games (H2), with replication rules for two more agents (H3).

3 2. Background and related work

- **Neuron-level interpretability.**
  - Circuits (Olah et al. 2020).
  - Polysemanticity and superposition, which motivate dictionary learning (Bricken et al. 2023).
  - Goodfire's counter-argument that neurons are mostly clean (2026).
- **RL interpretability.**
  - An Atari model zoo with activation analysis (Such et al. 2019).
  - CoinRun attribution and model editing (Hilton et al. 2020).
  - Probe-based steering of planning in Sokoban (Bush et al. 2025).
- **Ground truth for Atari.**
  - AtariARI RAM annotations (Anand et al. 2019).
  - OCAtari object extraction (Delfosse et al. 2023).
- **Evaluation protocol.** Sticky actions (Machado et al. 2018).

4 3. Method

4.1 3.1 Agents and fidelity

- **Agents.** `edbeeching/atari_2B_atari_mspacman_{1111,2222,3333}`:
  - Sample Factory 2.0 APPO, 2B env steps, clipped reward, $\gamma = 0.99$, no RNN;
  - Nature-CNN torso (conv 32/64/64) → fc 512 ReLU → linear policy (9 actions) and value heads.
- **Environment.** The wrapper chain is `MsPacmanNoFrameskip-v4` → no-op reset → max-and-skip 4 → episodic life → reward clipping → 84×84 grayscale → stack 4.
- **Fidelity.** We reproduce the wrapper chain and network in PyTorch and verify both against a golden trajectory generated with sample-factory 2.0.3, gym 0.26.2 and ale-py 0.8.1:
  - 0 mismatched observations in 446 steps;
  - max logit difference 1.1e-05.

  Appendix A has the details.

4.2 3.2 Ground truth

- **RAM decoding** follows OCAtari and AtariARI.
- **Frightened mode** was identified by correlating RAM bytes with blue ghost pixels and power-pill events.
- **Maze.** A 14×18 lattice graph built from the frame gives breadth-first-search distances and directions.
- **Features.** 40 features in five families (Appendix B). Steps where Ms. Pac-Man does not move (death animation, ready screen, level transition) are excluded.

4.3 3.3 Data and stochasticity

- **Splits.** 100 discovery games (seeds 0–99) and 100 held-out games (seeds 10000–10099) per agent.
- **Action sampling.** Actions are sampled from the policy with a per-game RNG, so steered and baseline games share random numbers.

- **Stochasticity fallback.** A preregistered check found that without sticky actions, same-start games stayed identical for a median of 1230 steps. The preregistered fallback (sticky actions, p = 0.25) was therefore used for all data (median divergence 320 steps).

4.4 3.4 Audit (H1)

- **Per-unit score.** For each alive fc unit (positive pole = post-ReLU activation):
  - R^2_1, the out-of-sample R^2 of 20-bin means of each single feature;
  - R^2_all, the out-of-sample R^2 of gradient boosting on all features.
- **One-job criterion.** R^2_all ≥ 0.3 and max R^2_1 ≥ 0.7 · R^2_all.
- **Controls.**
  - (a) random orthonormal directions;
  - (b) non-negative mixtures of 2–8 units;
  - (c) an untrained, orthogonally initialised network on the same states.
- **Test.** Game-cluster bootstrap of the difference in one-job fraction. H1 is supported if the 95% CI against (a) excludes 0.

4.5 3.5 Steering (H2) and replication (H3)

- **Why intervene upstream of fc.** Because the heads are linear and there is no recurrence:
  - clamping an fc unit only adds a constant action bias;
  - ablating it removes a state-dependent vote;
  - conv3 interventions act through 512 ReLUs.
- **Candidate selection** (discovery data only):
  1. Exact open-loop screens, using a rank-1 update for single conv3 units.
  2. Closed-loop pilots on discovery seeds.
  3. Two candidates frozen in a committed addendum.
- **Confirmatory design** (held-out seeds):
  - 100 baseline games, 50 games per candidate, and 20 games per null unit;
  - nulls: 99 random alive fc units, or the 63 other conv3 channels at the same cell;
  - pass criteria: predicted sign with Holm-corrected paired sign-flip p < 0.05, **and** an effect beyond the 95th percentile of the null.
- **Replication (H3).** Deterministic selection rules transfer the procedure to agents 2222 and 3333.

5 4. Results

5.1 4.1 Fidelity and ground truth

check	value
golden trajectory steps (incl. 2 life losses and game over)	446
steps with any observation pixel mismatch	0
max abs. logit difference	1.1e-05
max abs. value difference	4.8e-07
reference stack	sample-factory 2.0.3, gym 0.26.2, ale-py 0.8.1, torch 1.13.1 (WSL)

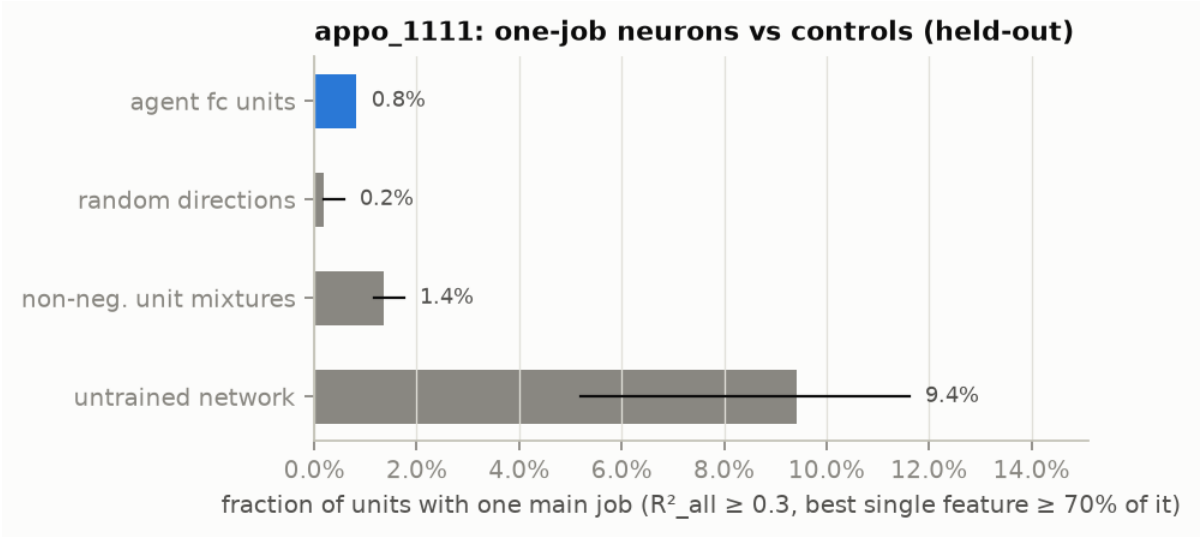
validation (discovery games, sticky actions)	value
agent steps checked	161995 (100 games)
RAM edible bit (bit 7 of ram[1+i]) vs blue sprite pixels, per ghost-step	97.1%
frightened timer ram[116] > 0 vs any ghost edible	100.0%

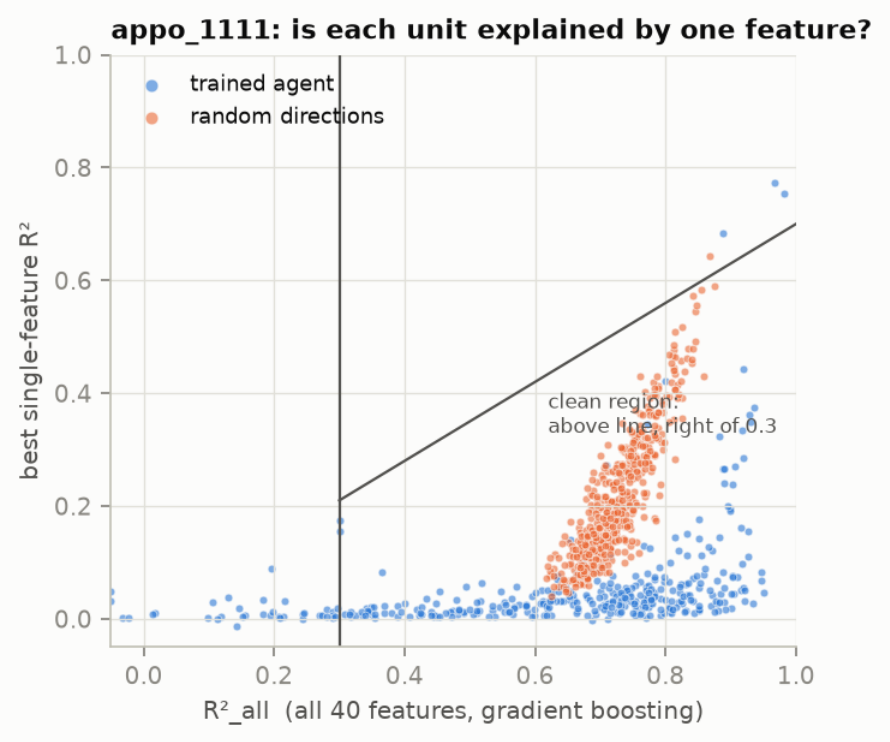
validation (discovery games, sticky actions)	value
eaten pellets at Ms. Pac-Man's lattice node (26825 events)	100.0%
steps in control (moving; excludes death, ready and level transitions)	85.1%
distinct maze layouts seen	4

5.2 4.2 H1: one-job neurons

unit set (held-out, confirmatory)	units	one-job fraction [95% CI]	median R^2_all	median best single R^2
trained agent (fc units)	367	0.8% [0.8%, 0.8%]	0.708	0.031
random directions (control a)	512	0.2% [0.2%, 0.6%]	0.719	0.185
non-negative unit mixtures (control b)	512	1.4% [1.2%, 1.8%]	0.720	0.053
untrained network (control c)	499	9.4% [5.2%, 11.6%]	0.537	0.096

H1 test (agent minus random directions): 0.6% [0.2%, 0.6%]; H1 supported.





Exploratory conjunctions (not preregistered):

explanatory variable(s) (exploratory, post hoc)	median R^2	units with $R^2 > 0.3$	units with $R^2 > 0.5$
place (lattice node)	0.127	11.7%	2.5%
place x heading	0.202	30.5%	7.4%
place x frightened	0.165	23.4%	6.0%
pellets left (20 bins) x frightened	0.022	2.7%	1.1%
danger distance (20 bins) x danger direction	0.018	0.3%	0.0%

5.3 4.3 H2: steering

Off-target effects:

Discovery-phase evidence that led to the candidates (exploratory):

rank	conv3 channel @ cell	Δ P(move toward cell) at junctions	dose (p99)
repel 1	ch 18 @ (5, 6)	-5.8 pp	24.6
repel 2	ch 15 @ (0, 6)	-5.1 pp	27.2
repel 3	ch 40 @ (4, 6)	-4.0 pp	34.8
repel 4	ch 36 @ (5, 6)	-3.6 pp	22.7
repel 5	ch 58 @ (5, 6)	-3.6 pp	18.4
attract 1	ch 16 @ (6, 2)	+3.6 pp	23.0
attract 2	ch 8 @ (1, 0)	+3.0 pp	21.6
attract 3	ch 38 @ (5, 0)	+3.0 pp	20.0
attract 4	ch 34 @ (6, 0)	+2.7 pp	28.1
attract 5	ch 46 @ (6, 0)	+2.6 pp	14.5

intervention (discovery pilot, paired games)	targeted metric	baseline → steered	Δ [95% CI]	Δ return	Δ deaths / 1k
c3_ch20_ablate	deaths_per_1k	2.536 → 2.730	0.194 [-0.183, 0.671]	-210	0.19
c3_ch20_clamp_frightened	ghosts_per_pill	0.392 → 0.213	-0.179 [-0.292, -0.060]	-3668	5.28
c3_ch20_clamp	deaths_per_1k	2.536 → 9.071	6.535 [5.265, 7.759]	-4047	6.54
c3_ch39_ablate	deaths_per_1k	2.536 → 4.169	1.633 [0.646, 2.502]	-2566	1.63
c3_ch53_ablate	deaths_per_1k	2.536 → 2.665	0.129 [-0.035, 0.305]	-229	0.13
c3_ch18_cell56	approach	0.430 → 0.375	-0.056 [-0.068, -0.044]	-3520	5.57
fc378_ablate	action_given_trigger	0.502 → 0.475	-0.027 [-0.055, -0.000]	-1307	0.67
fc453_ablate	action_given_trigger	0.483 → 0.298	-0.184 [-0.204, -0.164]	-2739	1.52
fc188_clamp	action_given_trigger	0.384 → 0.302	-0.082 [-0.091, -0.072]	-3047	1.43

5.4 4.4 H3: replication

6 5. Discussion

(Written after all confirmatory runs; see section 9.)

7 6. Limitations

- **Feature dictionary.** The one-job criterion depends on a finite 40-feature dictionary of *single* variables, so conjunctive codes are counted as "not clean" by construction. The conjunction analysis is post hoc.
- **Sticky actions.** These change the agent's task (it was trained without them) and lower its score. All results describe the agent under p = 0.25.
- **Dependence between games.** Games that share a start state overlap for their first few hundred steps. We report a start-state-cluster bootstrap as a robustness check.
- **Scope of interventions.** Steering was tested at fixed doses (discovery p99) and only for the frozen intervention types; dose-response is exploratory.
- **Agents.** Three public agents of one architecture and algorithm. Generalisation to other algorithms (DQN, PPO with IMPALA-ResNet) is untested.

8 7. Conclusion

(Written after all confirmatory runs.)

9 Appendix A. Fidelity details

See `reports/01_fidelity.md`.

10 Appendix B. Feature list

See `src/pacneuron/features.py` and `reports/02_ram_ground_truth.md`.

11 Appendix C. Preregistration and deviations

See `reports/preregistration.md`, `reports/preregistration_addendum.md` and `reports/00_summary.md`.

12 AI statement

This codebase and paper were written by Claude Code (Anthropic's Claude), following an exact specification from a human author who directed the research. Human cleanup and rewriting are in progress.